

MILAN VOJNOVIĆ

Professor of Data Science · Head, Department of Statistics, LSE · Visiting Scientist, Meta

m.vojnovic@lse.ac.uk | Department of Statistics, Columbia House, Office 8.04, Houghton Street, London WC2A 2AE, UK

<https://www.lse.ac.uk/people/milan-vojnovic> | <https://milanvojnovic.github.io/>

AREAS OF INTEREST

Artificial intelligence, AI alignment, inference-scaling, machine learning, machine learning optimisation, multi-agent systems, multi-armed bandits, optimisation, pre-/post-training, ranking algorithms, reinforcement learning, and reinforcement learning from human feedback.

EDUCATION

Ph.D., Technical Sciences

2003

École Polytechnique Fédérale de Lausanne (EPFL), Switzerland

Nominated for EPFL Best Ph.D. Thesis Award.

M.Sc., Electrical Engineering

1998

University of Split, Croatia

B.Sc., Electrical Engineering

1995

University of Split, Croatia

EXPERIENCE

Professor, Chair in Data Science

Oct 2016 –

London School of Economics (LSE), Department of Statistics

- ▶ Head of Department, Sept 2025–
- ▶ Knowledge Exchange and Impact Coordinator, Sept 2022–2024, 2025–2026
- ▶ Faculty Affiliate, LSE Data Science Institute, Sept 2024–
- ▶ Programme Director for MSc Data Science, Sept 2017 – Sept 2022

Visiting Scientist

Apr 2025–June 2026 | Apr 2022 – Feb 2024 | Jun 2019 – Apr 2021

Meta — Central Applied Science / Core Data Science

Visiting Researcher

Jan – Feb 2024

KAIST, Kim Jaechul Graduate School of AI, Seoul, Republic of Korea

Senior Researcher

Mar 2010 – Sept 2016

Researcher

May 2005 – Mar 2010

Associate Researcher	<i>Jan 2004 – May 2005</i>
<i>Microsoft Research — Systems and Networking; Machine Learning and Perception</i>	
Affiliated Lecturer	<i>Sept 2014 – Jun 2016</i>
<i>University of Cambridge, Statistical Laboratory</i>	
Visiting Professor	<i>Sept 2005 – Jun 2013</i>
<i>University of Split, Department of Computer Science</i>	
Research Assistant	<i>Sept 1999 – Jun 2003</i>
<i>EPFL</i>	
Research Intern	<i>Aug – Oct 2001</i>
<i>Bell Laboratories, Mathematical Research Center, Murray Hill, New Jersey</i>	
Graduate Student	<i>Sept 1998 – Jun 1999</i>
<i>EPFL Pre-Doctoral School in Communication Systems</i>	
Graduate Student & Lecturer	<i>Sept 1995 – Jun 1998</i>
<i>University of Split</i>	

AWARDS

- 2026** ACM KDD Best Paper Award (Applied Data Science Track)
- 2022** LSE Excellence in Education Award
- 2020** Facebook Research Award – Systems for ML
- 2018** Criteo Faculty Research Award
- 2018** Huawei Technologies Research Grant
- 2015** Microsoft Achievement Award
- 2010** ACM SIGMETRICS Rising Star Researcher Award
- 2007** IEEE IWQoS Best Student Paper Award
- 2005** ERCIM Cor Baayen Award
- 2005** IEEE INFOCOM Best Paper Award
- 2005** ACM SIGMETRICS Best Paper Award
- 2003** EPFL Best Ph.D. Thesis Nomination
- 2001** ITC-17 Best Student Paper Award
- 1999** EPFL Best Student Ph.D. Research Fellowship
- 1995** Best Student Award, University of Split

PUBLICATIONS

Books

- Milan Vojnović. **Contest Theory: Incentive Mechanisms and Ranking Methods.** *Cambridge University Press, Feb 2016.*

Research Papers

- ▶ Kim J. and Vojnović M. **Learning to Schedule in Parallel-Server Queues with Stochastic Bilinear Rewards.** *IEEE Trans. on Networking*, July 2026.
- ▶ Haimovich D., Linder F., Perini L., Tax N., Vojnović M. **On the Convergence of Multicalibration Gradient Boosting.** *arXiv:2602.06773*, 2026.
- ▶ Perini L., Haimovich D., Linder F., Tax N., Karamshuk D., Vojnović M., Okati N. and Apostolopoulos P.A. **MCGrad: Multicalibration at Web Scale.** *ACM KDD (Applied Data Science Track)*, 2026.
- ▶ Lee J., Jang K., Jun K., Vojnović M. and Yun S. **GL-LowPopArt: A Nearly Instance-Wise Minimax-Optimal Estimator for Generalized Low-Rank Trace Regression.** *AISTATS*, 2026.
- ▶ Vojnović M. and Wang Y. **Sketching Stochastic Valuation Functions.** *INFORMS Journal on Optimization*, April 2026.
- ▶ Kim J., Vojnović M. and Oh M. **Oracle-Efficient Combinatorial Semi-Bandits.** *NeurIPS*, 2025.
- ▶ Vojnović M. and Yun S. **What is the Alignment Objective of GRPO?** *arXiv:2502.18548*, 2025.
- ▶ Kim J., Vojnović M. and Yun S. **An Adaptive Approach for Infinitely Many-armed Bandits under Generalized Rotting Constraints.** *NeurIPS*, 2024.
- ▶ Haimovich D., Karamshuk D., Linder F., Tax N., and Vojnović M. **On the Convergence of Loss and Uncertainty-based Active Learning Algorithms.** *NeurIPS*, 2024.
- ▶ Wang Y., Chen W. and Vojnović M. **Combinatorial Bandits for Maximum Value Reward Function under Maximum Value and Index Feedback.** *ICLR*, 2024.
- ▶ Yi J. and Vojnović M. **Doubly Adversarial Federated Bandits.** *ICML*, 2023.
- ▶ Yi J. and Vojnović M. **On Regret-optimal Cooperative Nonstochastic Multi-armed Bandits.** *AAMAS*, 2023.
- ▶ Kim J., Vojnović M. and Yun S. **Rotting Infinitely Many-armed Bandits.** *ICML*, 2022.
- ▶ Kim J. and Vojnović M. **Scheduling Servers with Stochastic Bilinear Rewards.** *arXiv*, Jun 2022.
- ▶ Vojnović M. and Zhou K. **Dynamics and Inference for Voter Model Processes.** *arXiv*, Nov 2022.
- ▶ Vojnović M., Yun S. and Zhou K. **Accelerated MM Algorithms for Inference of Ranking Scores from Comparison Data.** *Operations Research*, 2022.
- ▶ Haimovich D., Karamshuk D., Leeper T., Riabenko E. and Vojnović M. **Popularity Prediction for Social Media over Arbitrary Time Horizons.** *VLDB*, 2022.
- ▶ Lee D., Vojnović M. and Yun S. **Test Score Algorithms for Budgeted Stochastic Utility Maximization.** *INFORMS Journal on Optimization*, 2022.
- ▶ Lee D. and Vojnović M. **Scheduling Jobs with Stochastic Holding Costs.** *arXiv*, Sept 2022.
- ▶ Lee D. and Vojnović M. **Scheduling Jobs with Stochastic Holding Costs.** *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- ▶ Sekar S., Vojnović M. and Yun S. **A Test Score-Based Approach to Stochastic Submodular Optimization.** *Management Science*, Vol. 67, No. 7, pp. 1075–1092, 2021.
- ▶ Sentenac F., Yi J., Calauzènes, Perchet V. and Vojnović M. **Pure Exploration and Regret Minimization in Matching Bandits.** *ICML*, 2021.
- ▶ Huang Z., Radunović B., Vojnović M. and Zhang Q. **Communication Complexity of Approximate Maximum Matching in the Message-Passing Model.** *Distributed Computing*, Vol. 33, No. 66, pp. 515–531, 2020.
- ▶ Vojnović M., Yun S. and Zhou K. **Convergence Rates of Gradient Descent and MM Algorithms for Bradley-Terry Models.** *AISTATS*, 2020.
- ▶ Shah V., Gulikers L., Massoulié L. and Vojnović M. **Adaptive Matching for Expert Systems with Uncertain Task Types.** *Operations Research*, Vol. 68, No. 5, pp. 1403–1424, 2020.

- ▶ Draief M., Kutzkov K., Scaman K. and Vojnović M. **KONG: Kernels for Ordered-Neighborhood Graphs.** *NeurIPS (Spotlight)*, 2018.
- ▶ Alistarh D., Grubic D., Li J., Tomioka R. and Vojnović M. **Communication-Efficient Stochastic Gradient Descent with Applications to Neural Networks.** *NeurIPS (Spotlight)*, 2017.
- ▶ Shah V., Gulikers L., Massoulié L. and Vojnović M. **Adaptive Matching for Expert Systems with Uncertain Task Types.** *Allerton Conference*, 2017.
- ▶ Vojnović M. **Contest Theory.** *Communications of the ACM*, May 2017.
- ▶ Alistarh D., Li J., Tomioka R. and Vojnović M. **Quantized Stochastic Gradient Descent.** *NIPS Workshop on Optimisation for Machine Learning (OPT)*, 2016.
- ▶ Fogel F., d'Aspremont A. and Vojnović M. **Spectral Ranking using Seriation.** *Journal of Machine Learning Research (JMLR)*, Vol. 17, No. 1, pp. 3013–3057, Jan 2016.
- ▶ Vojnović M. and Yun S. **Parameter Estimation for Generalized Thurstone Choice Models.** *ICML*, pp. 498–506, 2016.
- ▶ Rasley J., Karanasos K., Kandula S., Fonseca R., Vojnović M. and Rao S. **Efficient Queue Management for Cluster Scheduling.** *EuroSys*, 2016.
- ▶ Alistarh D., Iglesias J. and Vojnović M. **Streaming Min-Max Graph Partitioning.** *NeurIPS*, 2015.
- ▶ Alistarh D., Gelashvili R. and Vojnović M. **Fast and Exact Majority Population Protocol.** *ACM PODC*, pp. 47–56, 2015.
- ▶ Alistarh D., Gelashvili R. and Vojnović M. **Lock-Free Algorithms under Stochastic Schedulers.** *ACM PODC*, pp. 251–260, 2015.
- ▶ Huang Z., Radunović B., Vojnović M. and Zhang Q. **Communication Complexity of Approximate Matching in Distributed Graphs.** *STACS*, pp. 460–473, 2015.
- ▶ Zong B., Gkantsidis C. and Vojnović M. **Herdin Small Streaming Queries.** *ACM DEBS*, pp. 46–57, 2015.
- ▶ Bourse F., Lelarge M. and Vojnović M. **Balanced Graph Edge Partition.** *ACM KDD*, pp. 1456–1465, 2014.
- ▶ Fogel F., d'Aspremont A. and Vojnović M. **SerialRank: Spectral Ranking using Seriation.** *NeurIPS*, pp. 900–908, 2014.
- ▶ Bachrach Y., Syrgkanis V., Tardos É. and Vojnović M. **Strong Price of Anarchy, Utility Games and Coalitional Dynamics.** *SAGT*, pp. 218–230, 2014.
- ▶ Tsourakais C.E., Gkantsidis C., Radunović B. and Vojnović M. **Fennel: Streaming Graph Partitioning for Massive Scale Graphs.** *ACM WSDM*, pp. 33–342, 2014.
- ▶ Bachrach Y., Syrgkanis V. and Vojnović M. **Incentives and Efficiency in Uncertain Collaborative Environments.** *WINE*, pp. 26–39, 2013.
- ▶ Draief M. and Vojnović M. **Convergence Speed of Binary Interval Consensus.** *SIAM Journal on Control and Optimization*, Vol. 5, No. 3, pp. 1087–1109, 2012.
- ▶ Jung K., Kim B.Y. and Vojnović M. **Distributed Ranking in Networks with Limited Memory and Communication.** *IEEE ISIT*, pp. 980–984, 2012.
- ▶ Liu Z., Radunović B. and Vojnović M. **Continuous Distributed Counting for Non-Monotonic Streams.** *ACM PODS*, pp. 307–318, 2012.
- ▶ Draief M. and Vojnović M. **Bargaining Dynamics in Exchange Networks.** *Allerton Conference*, 2010.
- ▶ Vojnović M. and Proutiere A. **Hop Limited Flooding over Dynamic Networks.** *IEEE INFOCOM*, pp. 685–693, 2011.
- ▶ Gunawardena D., Karagiannis T., Proutiere A., Santos-Neto E. and Vojnović M. **Scoop: Decentralized and Opportunistic Multicasting of Information Streams.** *ACM MobiCom*, pp. 169–180, 2011.
- ▶ Nguyen T. and Vojnović M. **Weighted Proportional Allocation.** *ACM SIGMETRICS*, pp. 173–184, 2011.
- ▶ Karagiannis T., Le Boudec J.-Y. and Vojnović M. **Power Law and Exponential Decay of**

Interconnect Times between Mobile Devices. *IEEE Trans. on Mobile Computing*, Vol. 9, No. 10, pp. 1377–1390, 2010.

- ▶ Vojnović M., Gupta V., Karagiannis T. and Gkantsidis C. **Sampling Strategies for Epidemic-Style Information Dissemination.** *IEEE/ACM Transactions on Networking*, Vol. 18, No. 4, pp. 1013–1025, 2010.
- ▶ Hu L., Le Boudec J.-Y. and Vojnović M. **Optimal Channel Choice for Collaborative Ad-Hoc Dissemination.** *IEEE INFOCOM*, pp. 614–622, 2010.
- ▶ Draief M. and Vojnović M. **Convergence Speed of Binary Interval Consensus.** *IEEE INFOCOM*, pp. 1792–1800, 2010.
- ▶ Vojnović M., Cruise J., Gunawardena D. and Marbach P. **Ranking and Suggesting Popular Items.** *IEEE Trans. on Knowledge and Data Engineering*, Vol. 21, No. 8, pp. 1133–1146, 2009.
- ▶ Robertson S., Vojnović M. and Weber I. **Rethinking the ESP Game.** *CHI Extended Abstracts*, pp. 3937–3942, 2009.
- ▶ Gunawardena D., Karagiannis T., Proutiere A. and Vojnović M. **Characterizing Podcast Services: Publishing, Usage, and Dissemination.** *ACM SIGCOMM IMC*, pp. 209–222, 2009.
- ▶ Perron E., Vasudevan D. and Vojnović M. **Using Three States for Binary Consensus on Complete Graphs.** *IEEE INFOCOM*, pp. 2527–2535, 2009.
- ▶ DiPalantino D. and Vojnović M. **Crowdsourcing and All-Pay Auctions.** *ACM EC*, pp. 119–128, 2009.
- ▶ Karagiannis T. and Vojnović M. **Behavioural Profiles for Advanced Email Features.** *WWW*, pp. 711–720, 2009.
- ▶ Kesidis G., Hamadeh I., Jin O., Jiwasurat S. and Vojnović M. **A Model of the Spread of Randomly Scanning Internet Worms that Saturate Access Links.** *ACM TOMACS*, Vol. 18, No. 2, 2008.
- ▶ Massoulié L. and Vojnović M. **Coupon Replication Systems.** *IEEE/ACM Transactions on Networking*, Vol. 16, No. 3, pp. 603–616, 2008.
- ▶ Vojnović M. and Ganesh A.J. **On the Race of Worms, Alerts, and Patches.** *IEEE/ACM Transactions on Networking*, Vol. 16, No. 3, 2008.
- ▶ Suchanek F.M., Vojnović M. and Gunawardena D. **Social Tags: Meaning and Suggestions.** *ACM CIKM*, pp. 223–232, 2008.
- ▶ Vojnović M., Gupta V., Karagiannis T. and Gkantsidis C. **Sampling Strategies for Epidemic-Style Information Dissemination.** *IEEE INFOCOM*, pp. 1678–1686, 2008.
- ▶ Liu S., Vojnović M. and Gunawardena D. **Competitive and Considerate Congestion Control for Bulk Data Transfers.** *IEEE IWQoS*, pp. 1–9, 2007.
- ▶ Karagiannis T., Le Boudec J.-Y. and Vojnović M. **Power Law and Exponential Decay of Inter Contact Times between Mobile Devices.** *ACM MobiCom*, pp. 183–194, 2007.
- ▶ Le Boudec J.-Y. and Vojnović M. **The Random Trip Model: Stability, Stationary Regime, and Perfect Simulation.** *IEEE/ACM Transactions on Networking*, Vol. 16, No. 6, pp. 1153–1166, 2006.
- ▶ Altman E., Barman D., Tuffin B. and Vojnović M. **Parallel TCP Sockets: Simple Model, Throughput and Validation.** *IEEE INFOCOM*, 2006.
- ▶ Gkantsidis C., Karagiannis T., Rodriguez P. and Vojnović M. **Planet Scale Software Updates.** *ACM SIGCOMM*, pp. 423–434, 2006.
- ▶ Vojnović M. and Le Boudec J.-Y. **On the Long-Run Behaviour of Equation-Based Rate Control.** *IEEE/ACM Transactions on Networking*, Vol. 13, No. 3, pp. 568–581, 2005.
- ▶ PalChaudhuri S., Le Boudec J.-Y. and Vojnović M. **Perfect Simulations for Random Trip Mobility Models.** *Annual Simulation Symposium*, pp. 72–79, 2005.
- ▶ Key P., Massoulié L. and Vojnović M. **Farsighted Users Harness Network Time-Diversity.** *IEEE INFOCOM*, pp. 2383–2394, 2005.
- ▶ Le Boudec J.-Y. and Vojnović M. **Perfect Simulation and Stationary of a Class of Mobility Models.** *IEEE INFOCOM*, pp. 2743–2754, 2005.

- ▶ Massoulié L. and Vojnović M. **Coupon Replication Systems.** *ACM SIGMETRICS*, pp. 2–13, 2005.
- ▶ Vojnović M. and Ganesh A.J. **On the Effectiveness of Automatic Patching.** *ACM Workshop on Rapid Malcode (WORM)*, pp. 41–50, 2005.
- ▶ Andrews M. and Vojnović M. **Scheduling Reserved Traffic in Input-Queued Switches: New Delay Bounds via Probabilistic Techniques.** *IEEE Journal on Selected Areas in Communications*, Vol. 21, No. 4, pp. 595–605, 2003.
- ▶ Andrews M. and Vojnović M. **Scheduling Reserved Traffic in Input-Queued Switches: New Delay Bounds via Probabilistic Techniques.** *IEEE INFOCOM*, 2003.
- ▶ Vojnović M. and Le Boudec J.-Y. **Bounds for Independent Regulated Inputs Multiplexed in a Service Curve Network Element.** *IEEE Transactions on Communications*, Vol. 51, No. 5, pp. 735–740, 2003.
- ▶ Vojnović M. and Le Boudec J.-Y. **Stochastic Bound on Delay for Guaranteed Rate Nodes.** *IEEE Communications Letters*, Vol. 6, No. 10, pp. 449–451, 2002.
- ▶ Vojnović M. and Le Boudec J.-Y. **Stochastic Analysis of Some Expedited Forwarding Networks.** *IEEE INFOCOM*, 2002.
- ▶ Vojnović M. and Le Boudec J.-Y. **On the Long-Run Behaviour of Equation-Based Rate Control.** *ACM SIGCOMM*, pp. 103–116, 2002.
- ▶ Vojnović M. and Le Boudec J.-Y. **Bounds for Independent Regulated Inputs Multiplexed in a Service Curve Network Element.** *GLOBECOM*, pp. 1857–1861, 2001.
- ▶ Vojnović M. and Rožić N. **An Evaluation of the ABR Explicit-Rate Allocation Interfering with the Guaranteed Services Traffic.** *Computer Networks*, Vol. 34, No. 1, pp. 139–155, 2000.
- ▶ Vojnović M., Le Boudec J.-Y. and Boutremans C. **Global Fairness of Additive-Increase and Multiplicative-Decrease with Heterogeneous Round-Trip Times.** *IEEE INFOCOM*, pp. 1303–1312, 2000.
- ▶ Hubaux J.-P., Le Boudec J.-Y., Giordano S., Hamdi M., Blažević L., Buttán L. and Vojnović M. **Towards Mobile Ad-Hoc WANS: Terminodes.** *IEEE WCNC*, pp. 1052–1059, 2000.

PATENTS

- ▶ Apparatuses and Methods for Detecting Malware, Huawei Technologies Co., LTD., US patent application No. 86216338US04, Nov 2021
- ▶ Efficient Queue Management for Cluster Scheduling, Microsoft Corporation, patent application, Apr 2017
- ▶ Quantized Stochastic Gradient Descent, Microsoft Corporation, patent application, May 2016
- ▶ Decentralized Relaying Algorithm for Mobile Devices, Microsoft Corporation, Patent No. 8,886,731, granted Nov 11, 2014
- ▶ Allocating Tasks to Machines in Computing Clusters, Microsoft Corporation, Patent No. 9,695,009, granted Apr 8, 2014
- ▶ Optimal Query Assignment: Network Load Minimization, Microsoft Corporation, Patent filed Mar 3, 2014
- ▶ Ranking System, Microsoft Corporation, Patent No. 8,478,762, granted Jul 2, 2013
- ▶ Graph Partitioning for Massive Scale Graphs, Microsoft Corporation, Patent filed Apr 29, 2013
- ▶ Sampling Rules for Information Dissemination, Microsoft Corporation, Patent No. 8,081,581, granted Dec 20, 2011
- ▶ Modeling of Plurality of Contests at a Crowdsourcing Node, Microsoft Corporation, Patent No. 8,195,498, granted Jun 5, 2012
- ▶ Count Tracking in Distributed Environments, Microsoft Corporation, Patent filed Mar 15, 2012
- ▶ Ranking and Suggesting Candidate Objects, Microsoft Corporation, Patent No. 7,685,200, granted Mar 23, 2010

- ▶ Competitive and Considerate Congestion Control, Microsoft Corporation, Patent No. 7,649,841, granted Jan 19, 2010

TECHNOLOGY TRANSFER

Technology Transfers

- ▶ MCGrad: Multicalibration at Web Scale, Meta Platform, 2025
- ▶ Popularity Prediction Methods for a Community Integrity Platform, Facebook, 2021
- ▶ Network Stream Optimiser, Stream Processing Platform, Bing, 2014
- ▶ Tag Recommendation System, Microsoft Live Service, 2008
- ▶ 4CP: Background Transport Control Protocol, Microsoft Vista (beta), 2006

Consultancy, Demos and Prototypes

- ▶ “Office Delve Recommendation System,” consultancy and working with the product team, 2015
- ▶ “On the Algorithms that Underpin our Cloud,” poster, MSR TechFest 2015, Microsoft, Redmond, WA; with Dan Alistarh and Christos Gkantsidis
- ▶ “MSR Algorithms Virtual Center (VIRAL),” booth with demos, MSR TechFest 2014, Microsoft Redmond, WA; with Andrew Goldberg, Ravi Kanan, Yuval Peres, Madhu Sudan and Robert Tarjan
- ▶ “Streaming Partitioning for Massive-Scale Graphs,” demo, MSR TechFest 2013, Microsoft, Redmond, WA; with Christos Gkantsidis
- ▶ “Big Data Analytics and Infrastructure,” theme demo owner, MSR TechFest 2013, Microsoft, Redmond, WA — proposed the theme, coordinated presentation of 10 demos from across Microsoft Research
- ▶ “Sampling Based Methods for Range Partition,” Cosmos, Bing Infrastructure Team, 2010; with Fei Xu and Jingren Zhou
- ▶ “Ox: System for Large-Scale Information Dissemination in Mobile Networks,” demo, MSR TechFest 2009, Microsoft, Redmond, WA; with Thomas Karagiannis and Eliseu Santos-Neto
- ▶ “TagBooster: A System for Ranking and Suggesting Popular Items,” demo, MSR TechFest 2007, Microsoft, Redmond, WA; with Dinan Gunawardena
- ▶ “Farsighted Congestion Control,” demo, MSR TechFest 2006, Microsoft, Redmond, WA; with Peter Key and Laurent Massoulie

LECTURES

- ▶ From Empirical Success to Theoretical Foundations: Multicalibration Gradient Boosting at Scale, Invited Seminar, Department of Computer Science, University of Oxford, May 19, 2026
- ▶ Multicalibration at Web Scale, Invited Seminar, AI for Computing and Society, Max Planck Institute for Software Systems (MPI SWS), Jan 26, 2026
- ▶ On the Convergence of Loss and Uncertainty-Based Active Learning Algorithms, Invited Seminar, Seoul National University, Seoul, Republic of Korea, Feb 15, 2024
- ▶ Learning to Schedule, Invited Seminar, CS Colloquium, University of Warwick, UK, Oct 21, 2021
- ▶ Learning to Match in Online Platforms, Invited Seminar, Bristol Data Science Seminars, University of Bristol, UK, Feb 27, 2020
- ▶ Optimisation for Inference and Machine Learning, Plenary Talk, IMA and OR Society Conference on Mathematics of Operational Research, Aston University, Birmingham, UK, Apr 2019
- ▶ Scalable Methods for Machine Learning Optimisation, Invited Seminar, Statslab, University of Cambridge, UK, May 10, 2019
- ▶ Learning to Match in Online Platforms, Invited Seminar, Athens University of Economics and Business, Athens, Greece, Dec 12, 2019

- ▶ Scalable Inference for Statistical Models of Ranking Data, Invited Talk, Workshop on Challenges for Categorical Data Analysis, RWTH Aachen University, Germany, Oct 2018
- ▶ Scalable Machine Learning for Big Data Analytics, Invited Talk, Huawei European Research Symposium, Paris, France, Jan 2018
- ▶ Thurstone Choice Model Inference, Workshop on Next Generation Network Analytics, Royal Statistical Society, London, UK, Jan 2018
- ▶ On a Portfolio Selection Problem using Individual Test Scores, Invited Talk, Risk & Stochastics Conference, London, UK, Apr 2017
- ▶ How to Hire a Team using Individual Test Scores?, Invited Talk, LSE Maths Seminar on Combinatorics, Games and Optimisation, LSE, London, UK, Jan 2017
- ▶ Communication Efficient Stochastic Gradient Descent, Huawei Innovative Algorithmic and Mathematical Workshop on Next Generation Network Technology, Paris, France, Nov 2016
- ▶ Talk by an EDIC Alumnus, Invited Talk, EDIC Open House, EPFL, Switzerland, Mar 2016
- ▶ Majority Computation Problem, Invited Keynote Talk, MSR Theory Day, Redmond, WA, Mar 2016
- ▶ On Rating Skills and Hiring a Team, Invited Talk, Telefonica Research, Barcelona, Spain, Feb 2016
- ▶ Utility Sharing and Social Welfare, Invited Talk, LSE, London, UK, Jun 2015
- ▶ Utility Sharing and Social Welfare, Invited Talk, University of Cambridge, Department of Economics, Cambridge, UK, Jun 2015
- ▶ Talk by an EDIC Alumnus, Invited Talk, EDIC Open House, EPFL, Lausanne, Switzerland, Mar 2015
- ▶ A Spectral Algorithm for Ranking using Seriation, Invited Talk, Machine Learning in Cambridge, Cambridge, UK, Mar 2015
- ▶ Load Balancing Tasks with Overlapping Requirements, Invited Talk, Workshop on Community Detection, Institut Henri Poincaré, Paris, France, Feb 2015
- ▶ How to Divide Prize Money?, Invited Talk, Trinity Mathematical Society, Cambridge, UK, Feb 2015
- ▶ Crowdsourcing and All-Pay Auctions, Invited Talk, Contemporary Economic Issues, University of East Anglia, Norwich, UK, Nov 2014
- ▶ Consensus – A Retrospective, Invited Talk, Workshop on Performance and Control of Large-Scale Networks, EURANDOM, Eindhoven, Netherlands, Jun–Jul 2014
- ▶ Consensus – A Retrospective, Invited Talk, University of Oxford, UK, Jun 2014
- ▶ Fennel: Streaming Graph Partitioning for Massive Scale Graphs, Invited Talk, Workshop on Big Dynamic Distributed Data, VLDB 2013, Trento, Italy, 2013
- ▶ Balanced Graph Partitioning for Massive Scale Computations, Invited Talk, LIDS Seminar, MIT, Cambridge, MA, Dec 2013
- ▶ Balanced Graph Partitioning for Massive Scale Computations, Invited Talk, University of Warwick, UK, Dec 2013
- ▶ Cooperation and Efficiency in Utility Maximization Games, Invited Talk, Microsoft Research Silicon Valley, Theory Seminar, Oct 2013
- ▶ Efficiency and Redistribution of Welfare, Invited Talk, Hamilton Institute, Ireland, Feb 2013
- ▶ Continuous Distributed Counting for Non-Monotonic Streams, Invited Talk, INTIMATE “Big Data In Digital Life” workshop, Paris, France, Jun 2012
- ▶ Some Maths of Use in the Computer Systems World, Invited Talk, Cambridge Centre for Analysis, University of Cambridge, Oct 2012
- ▶ Consensus, Invited Talk, Lecture Series Algorithms and Complexity in Durham, University of Durham, UK, Oct 2012
- ▶ Sampling Based Range Partition Methods for Big Data Analytics, Invited Talk, INQUEST Workshop, University of Oxford, UK, Sept 2012

- ▶ Continuous Count Tracking for Distributed Non-Monotonic Streams, Invited Talk, NII Shonan Meeting: Large-Scale Distributed Computation, Shonan, Japan, Jan 2012
- ▶ Continuous Count Tracking for Distributed Non-Monotonic Streams, Invited Talk, Tokyo University of Technology, Japan, Jan 2012
- ▶ Revenue of Auctions: Truthful vs. Non-Truthful Designs, Invited Talk, INFORMS, Stockholm, Sweden, Jul 2011
- ▶ Weighted Proportional Allocation, Invited Talk, Microsoft Research Asia, Beijing, China, 2011
- ▶ Mechanism Design, Invited Talk, Summer Research Institute, EPFL, Switzerland, Jun 2011
- ▶ Computer Systems seen as Auctions, Keynote Talk, ACM SIGMETRICS 2010, New York, NY, Jun 2010
- ▶ Mechanism Design, Tutorial, Microsoft Research Tutorial Series, Cambridge, UK, Mar 2010
- ▶ Bargaining Dynamics in Exchange Networks, Invited Talk, Allerton, IL, Sept 2010
- ▶ Weighted Proportional Allocation, Invited Talk, University of Cambridge, UK, 2010
- ▶ Weighted Proportional Allocation, Invited Talk, Harvard University, MA, 2009
- ▶ Crowdsourcing and All-Pay Auctions, Invited Talk, University of California Berkeley, CA, Jul 2009
- ▶ Crowdsourcing and All-Pay Auctions, Invited Talk, Statslab Networks Seminar, University of Cambridge, UK, May 2009
- ▶ Binary Consensus Using Three States on Complete Graphs, Invited Talk, Epidemic Spreading and Networks, Imperial College London, UK, Oct 2008
- ▶ Consensus with Limited Memory and Signaling, Invited Talk, Statslab, University of Cambridge, UK, 2008
- ▶ Consensus with Limited Memory and Signaling, Invited Talk, NET-COOP 2008, Paris, France
- ▶ Social Networks Overview, Invited Talk, Unilever Spark Workshop, Norwich, UK, 2008
- ▶ Tagging done by YOU, Tutorial, Microsoft Research Tutorial Series, Cambridge, UK, 2008
- ▶ Power Law and Exponential Decay – Mobile Devices, Invited Talk, Hynet Colloquium, University of Maryland, MD, 2007
- ▶ Ranking and Suggesting Popular Items, Invited Talk, Workshop on Mathematics of Communication Networks, ENS, Paris, France, 2007
- ▶ Ranking and Suggesting Popular Items, Invited Talk, INFORMS, Eindhoven, Netherlands, 2007
- ▶ Coupon Replication Systems, Invited Talk, INFORMS, Pittsburgh, PA, 2006
- ▶ Random Trip Model, Three Lecture Mini-Course, Charles University Prague, Czech Republic, 2006
- ▶ Random Trip Mobility Models, Tutorial, ACM MobiCom 2006, with Jean-Yves Le Boudec, Los Angeles, CA, 2006
- ▶ Farsighted Congestion Control, Invited Talk, CSAIL, MIT, Cambridge, MA, 2005
- ▶ Planet Scale Software Updates, Invited Talk, Microsoft Research, Redmond, WA, 2005
- ▶ Planet Scale Software Updates, Invited Talk, TU Munich, Germany, 2005
- ▶ Random Trip Model, Invited Talk, Max Planck Institute für Informatik, Saarbrücken, Germany, 2005
- ▶ Software Updates, Tutorial, IFIP Performance 2005, with Ayalvadi Ganesh, Juan-les-Pins, France, 2005
- ▶ Coupon Replication Systems, Invited Talk, EPFL Summer Research Institute, Lausanne, Switzerland, 2004
- ▶ Random Trip Model, Invited Talk, University of Cambridge, UK, 2004
- ▶ Farsighted Congestion Control, Invited Talk, ARC-TCP Workshop, INRIA, Sophia-Antipolis, France, 2004
- ▶ Equation-Based Rate Control, Invited Talk, IPAM Workshop on Large Scale Communication Networks, UCLA, Los Angeles, CA, Mar 2002
- ▶ Equation-Based Rate Control, Invited Talk, Columbia University, New York City, NY, 2001
- ▶ Equation-Based Rate Control, Invited Talk, University of Massachusetts at Amherst, MA, 2001

COURSES TAUGHT

- ▶ **ST443 Machine Learning and Data Mining**, MSc in Data Science, LSE, Lecturer, 2024/25–2025/26. Topics: basics of statistical machine learning, linear classifiers, LDA, logistic regression, non-linear classifiers, QDA, Naive Bayes, k -NN classifiers, cross-validation, bootstrap, regularisation, trees and forests, boosting algorithms, SVMs, dimensionality reduction, and clustering algorithms.
- ▶ **ST456 Deep Learning**, MSc in Data Science, LSE, Lecturer, 2021/22–2022/23, 2024/25. Topics: single-layer networks and perceptron, deep neural networks, expressiveness of feedforward neural networks, training algorithms, CNNs, sequence models, RNNs, transformers, training language models, autoencoders, and generative adversarial networks.
- ▶ **ST510 Foundations of Machine Learning**, PhD Course, LSE, Lecturer, 2020/21–present. Topics: statistical learning theory, online optimisation, and neural networks.
- ▶ **ST449 Artificial Intelligence and Deep Learning**, MSc in Data Science, LSE, Lecturer, 2018/19–2020/21. Topics: neural networks, CNNs, RNNs, transformers, reinforcement learning, MDPs, dynamic programming, Monte Carlo methods, temporal difference methods, eligibility traces, function approximation, and policy gradient methods.
- ▶ **ST446 Distributed Computing for Big Data**, MSc in Data Science, LSE, Lecturer, 2017/18–2022/23. Topics: distributed file systems, key-value stores, MapReduce, Pregel, Spark RDDs, Spark SQL, Hive, HiveQL, Spark GraphX, stream processing, Spark Streaming, scalable ML, Spark MLlib, and distributed ML.
- ▶ **ST445 Managing and Visualising Data**, MSc in Data Science, LSE, Lecturer, 2017/18. Topics: exploratory data analysis, matrix data visualisation, ROC curves, AUC, PR curves, PCA, MDS, LLE, Isomap, Laplacian eigenmaps, SNE, t -SNE, and graph data visualisation.
- ▶ **Contest Theory**, Lent Term, Part III of the Mathematical Tripos, University of Cambridge, Lecturer, 2013–2015
- ▶ **Networked Systems Architecture**, University of Split, undergraduate computer science, Lecturer, 2005–2013
- ▶ Performance Evaluation, EPFL, master programme, Teaching Assistant, 2002
- ▶ Computer Networking, EPFL, undergraduate studies, Teaching Assistant, 2001
- ▶ Java Programming, EPFL, undergraduate studies, 2000
- ▶ Analogue and Digital Circuits, University of Split, undergraduate electrical engineering, laboratory and theoretical problem solving exercises, 1995–1998
- ▶ Antennas and Propagation, University of Split, undergraduate electrical engineering, laboratory and theoretical problem solving exercises, 1995–1998
- ▶ Numerical Methods in Electronics, University of Split, undergraduate electrical engineering, laboratory and theoretical problem solving exercises, 1995–1998

SUPERVISIONS

Ph.D. Thesis Advisor

- ▶ Hitesh Gudwani, LSE, 2025–
- ▶ Jialin Yi, LSE, 2018–2023. Now an Associate with Citadel, London, UK.
- ▶ Yiliu Wang, LSE, 2018–2022. Now a Researcher with Allen Institute, WA.
- ▶ Kaifang Zhou, LSE, 2017–2023. Now an LSE Fellow, Department of Statistics, LSE, London, UK.

Ph.D. Thesis Committee Member

- ▶ Nastaran Okati, *On Predictive Accuracy and Fairness in Human-AI Teams*, supervised by Manuel Gomez-Rodriguez, Max Planck Institute for Software Systems, Jul 2025
- ▶ Abheek Ghosh, *Contests: Equilibrium Analysis Design and Learning*, supervised by Paul Goldberg

and Edith Elkind, University of Oxford, Jun 2024

- ▶ Luke Marris, *Multiagent Training in N-Player General-Sum Games*, supervised by Thore Graepel and Jun Wang, UCL, Jan 2024
- ▶ Jee Hyun (Caleb) Park, *Leveraging Structural Properties of Product-Level Data of In-Store Field Experiments and Combinatorial Auctions*, supervised by Gah-Yi Band and Jeremie Gallien, London Business School, Dec 2023
- ▶ Tianlin Xu, *Generative Adversarial Networks for Sequential Learning*, supervised by Wicher Bergsma, LSE, London, UK, May 2022
- ▶ Rico Blaser, *Random Rotations in Machine Learning*, supervised by Piotr Fryzlewicz, LSE, London, UK, Dec 2021
- ▶ Lucas Maystre, *Efficient Learning from Comparisons*, supervised by Matthias Grossglauser, EPFL, Lausanne, Switzerland, Apr 2018
- ▶ Remi Varloot, *Dynamic Network Formation*, École Normale Supérieure, supervised by Laurent Massoulié, Paris, France, Jun 2018
- ▶ Fajwel Fogel, *Convex and Spectral Relaxations for Phase Retrieval, Seriation and Ranking*, supervised by Alexandre d'Aspremont and Francis Bach, École Polytechnique, Paris, France, Sept 2015
- ▶ Michal Ficek, *Tracking Users in Mobile Networks: Data Acquisition, Methods and their Limits*, supervised by Lukas Kencl, Czech Technical University in Prague, Czech Republic, Jun 2013

Post-docs

- ▶ Dabeen Lee, KAIST, 2021–2022. Now Assistant Professor at Seoul National University (SNU).
- ▶ Virag Shah, University of Texas at Austin, 2016–2017. Now with Uber.
- ▶ Se-Young Yun, KAIST, 2015–2016. Was a post-doc with Los Alamos National Laboratory, Los Alamos, New Mexico; now Associate Professor at KAIST.
- ▶ Dan Alistarh, EPFL, 2014–2015. Was a Researcher with MSR; now Professor at IST Austria (ISTA).

Interns

- ▶ Jung-Hun Kim, KAIST, South Korea, 2021
- ▶ Remi Jezequel, École Normale Supérieure, 2017
- ▶ Lucas Maystre, EPFL, 2016
- ▶ Jerry Li, MIT, 2016
- ▶ Hyun Jik Kim, University of Cambridge, 2015
- ▶ Gautam Kamath, MIT, 2015
- ▶ Fajwel Fogel, École Normale Supérieure, 2014
- ▶ Rati Gelashvili, MIT, 2014
- ▶ Jennifer Iglesias, CMU, 2014
- ▶ Florian Bourse, École Normale Supérieure, 2013
- ▶ Nan Li, University of California Santa Barbara, 2013
- ▶ Bo Zong, University of California Santa Barbara, 2013
- ▶ Charalampos Tsourakakis, CMU, 2012
- ▶ Alan Roytman, UCLA, 2012
- ▶ Zengfeng Huang, Hong Kong University of Science and Technology, 2012
- ▶ Zhenming Liu, Harvard University, 2011
- ▶ Vasilis Syrgkanis, Cornell University, 2011
- ▶ Jinshan Zhang, University of Liverpool, UK, 2011
- ▶ Amin Karabasi, EPFL, 2010
- ▶ Thanh Nguyen, Cornell University, 2009
- ▶ Dominic DiPalantino, Stanford University, 2008

- ▶ Elizeu Santos-Neto, University of British Columbia, 2008
- ▶ Dinkar Vasudevan, EPFL, 2008
- ▶ Fabian Suchanek, MPI, 2007
- ▶ James Cruise, University of Cambridge, 2007
- ▶ Varun Gupta, CMU, 2006
- ▶ Thomas Karagiannis, University of California Riverside, 2006
- ▶ Shao Liu, UIUC, 2005

Part III Master Programme Essays

- ▶ Aya Ahmed, “Allocating Tasks to Workers in Crowdsourcing Services,” 2015
- ▶ Adam Chudziak, “Statistical Inference and Optimal Auction Design,” 2015
- ▶ Benjamin Dubois, “Fair Allocation of Multi-Type Resources,” 2015
- ▶ Fikri Pitsuwan, “Statistical Inference and Optimal Auction Design,” 2015
- ▶ George Bignall, “Tug of War,” 2014

PROFESSIONAL SERVICE

Program Committee Activities

- ▶ Area Vice-Chair, Distributed, Parallel and P2P Data Management, IEEE ICDE 2019
- ▶ Co-Chair, Workshop on Machine Learning and User Decision Making, co-organised with Massoulie´ L., Paris, France, May 23–24, 2019
- ▶ Chair, Workshop on Data Science Theory and Practice, LSE, London, Mar 27–28, 2018
- ▶ Member, Performance 2018
- ▶ Member, Program Committee, ACM SIGMETRICS 2017
- ▶ Member, Program Committee, ACM SIGMETRICS 2016
- ▶ Member, Program Committee, ACM SIGMETRICS 2015
- ▶ Chair, Workshop on Algorithms and Data Science, Microsoft Research, Cambridge, UK, May 15, 2014; keynote speaker: Robert Tarjan
- ▶ Chair, Workshop on Big Data Analytics, Microsoft Research, Cambridge, UK, May 23–24, 2013; co-organised with Artur Czumaj and Jingren Zhou; keynote speakers: Surajit Chaudhuri, Sanjeev Khanna, and Gerhard Weikum
- ▶ Member, Program Committee, ACM SIGMETRICS 2014
- ▶ Member, Program Committee, ACM SIGMETRICS 2013
- ▶ Member, External Review Committee – Cloud Computing and Next-Generation Distributed Query Processing, ACM SIGMOD 2013
- ▶ Member, Program Committee, ACM SIGMETRICS 2012
- ▶ Member, Program Committee, ACM Electronic Commerce 2011
- ▶ Member, Program Committee, IFIP Performance 2011
- ▶ Member, Program Committee, ACM SIGMETRICS 2011
- ▶ Member, Program Committee, WWW 2011
- ▶ Member, Program Committee, IEEE INFOCOM 2011
- ▶ Chair, Workshop on the Economics of Networks, Systems, and Computation (NetEcon), 2010; co-organised with Mike Dahlin
- ▶ Chair, Workshop on the Future of Social Networking: Experts from Industry and Academia, Cambridge, UK, Nov 18, 2010; co-organised with Jon Crowcroft, Pietro Lio, Neil Walton, and Eiko Yoneki
- ▶ Member, Program Committee, IEEE INFOCOM 2010
- ▶ Member, Program Committee, WWW 2010
- ▶ Member, Program Committee, IEEE INFOCOM 2009

- ▶ Member, Program Committee, WWW 2009
- ▶ Member, Program Committee, IEEE INFOCOM 2008
- ▶ Chair, Workshop on Online Social Networks, Microsoft Research, Cambridge, UK, 2007; co-organised with Thomas Karagiannis, Peter Key, and Bala Krishnamurthy
- ▶ Member, Program Committee, IEEE INFOCOM 2007
- ▶ Member, Program Committee, ACM SIGCOMM 2007
- ▶ Member, Program Committee, IEEE INFOCOM 2006
- ▶ Chair, InterPerf 2006, ValueTools 2006; co-organised with Sean Collins

Steering Committees

- ▶ NetEcon – Workshop on the Economics of Networks, Systems and Computation, 2008–
- ▶ Workshop on Online Social Networks (WOSN), 2008–2011
- ▶ Incentive-Based Computing (IBC), 2006

Other Panels

- ▶ Member, SIGMETRICS Test of Time Award Committee, with Devavrat Shah (Chair) and Vijay Subramanian, 2025
- ▶ Evaluator, Horizon 2020, European Commission, Brussels, Belgium, Jun 12–22, 2017
- ▶ Evaluator, INRIA projects under the theme “Networks and Telecommunications,” Mar 22–24, 2016, Paris, France
- ▶ Member, Senior Leadership Team, Microsoft Research Algorithms Virtual Center (VIRAL), 2013–2016; other members included Nikhil Devanur, Andrew Goldberg, Ravi Kannan, Robert Kleinberg, Thomas Moscibroda, Yuval Peres, Robert Schapire, Madhu Sudan and Robert Tarjan
- ▶ Evaluator, FP7, European Commission, Brussels, Belgium, Dec 7–11, 2009

OTHER ACTIVITIES

Organiser of Lecture Series

- ▶ *Big Data Analytics*, MSR, Cambridge, UK, 2012. Speakers: Graham Cormode (AT&T Research), Tugkan Batu (LSE), Ely Porat (Bar-Ilan University), Tom Friedetzky (Durham University), Alexandre d’Aspremont (École Polytechnique, Paris), Nicolo Cesa-Bianchi (University of Milan) and Michael Benedikt (Oxford University).
- ▶ *Algorithms and Economics*, MSR, Cambridge, UK, 2011. Speakers: Kostas Bimpikis (MSR New England), Paul Goldberg (University of Liverpool), Michael Kearns (University of Pennsylvania), Robert Mataescu (MSR), Romans Pancs (University of Rochester), Harald Raecke (University of Warwick), Peyton Young (University of Oxford), Jenn Wortman Vaughan (UCLA), Richard Weber (University of Cambridge).